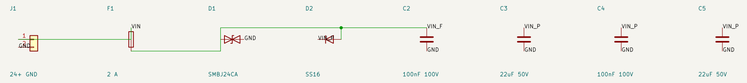
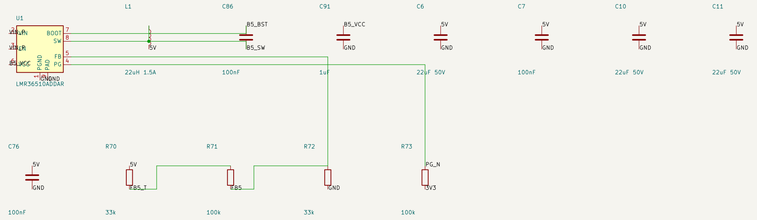
Power distribution, protection and local regulation | 55 components | 25 local nets

24 V INPUT, FUSE, CLAMP AND REVERSE-POLARITY PROTECTION



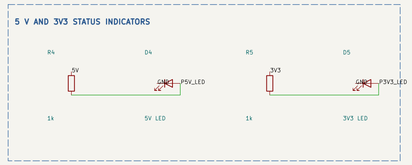
5 V BUCK CONVERTER, FEEDBACK AND OUTPUT CAPACITORS



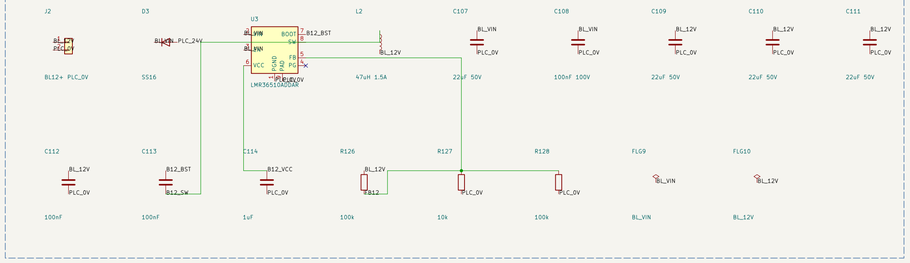
3V3 POINT-OF-LOAD REGULATOR AND ANALOG FILTER



5 V AND 3V3 STATUS INDICATORS



ISOLATED PLC-24 V TO 12 V BACKLIGHT CONVERTER



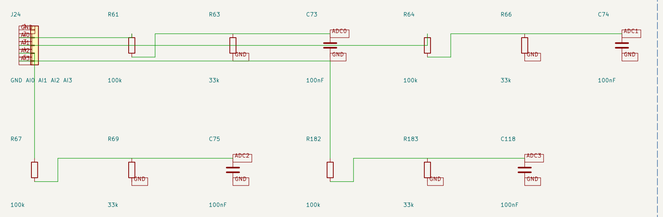
OFF-SHEET POWER AND GROUND REFERENCES



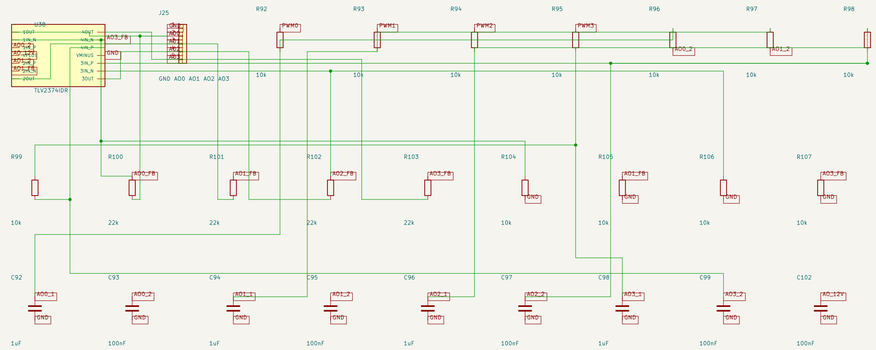
Method B: authoritative KiCad symbols grouped by power flow
LeanCam / uCNC schematic readability experiment
Sheet: /
File: lean-power-b.kicad_sch
Title: Power distribution, protection and local regulation
Size: A1 | Date: 2026-11-03 | Rev:
KiCad E.D.A. 10.0.1 | Id: 1/1

Analog inputs, 0–10 V outputs and 12 V rail | 43 components | 31 local nets

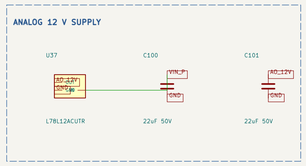
ANALOG INPUTS AI0–AI3



ANALOG OUTPUTS DO1–DO4



ANALOG 12 V SUPPLY



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

File: analog_io-python.kicad_sch

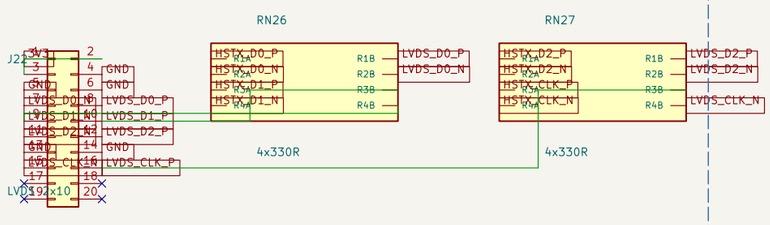
Title: Analog inputs, 0–10 V outputs and 12 V rail

Size: A1 | Date: 2026-11-03 | **Rev:**

KiCad E.D.A. 10.0.1 | Id: 1/1

LVDS transmitter, termination and connector | 3 components | 18 local nets

LVDS OUTPUT



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

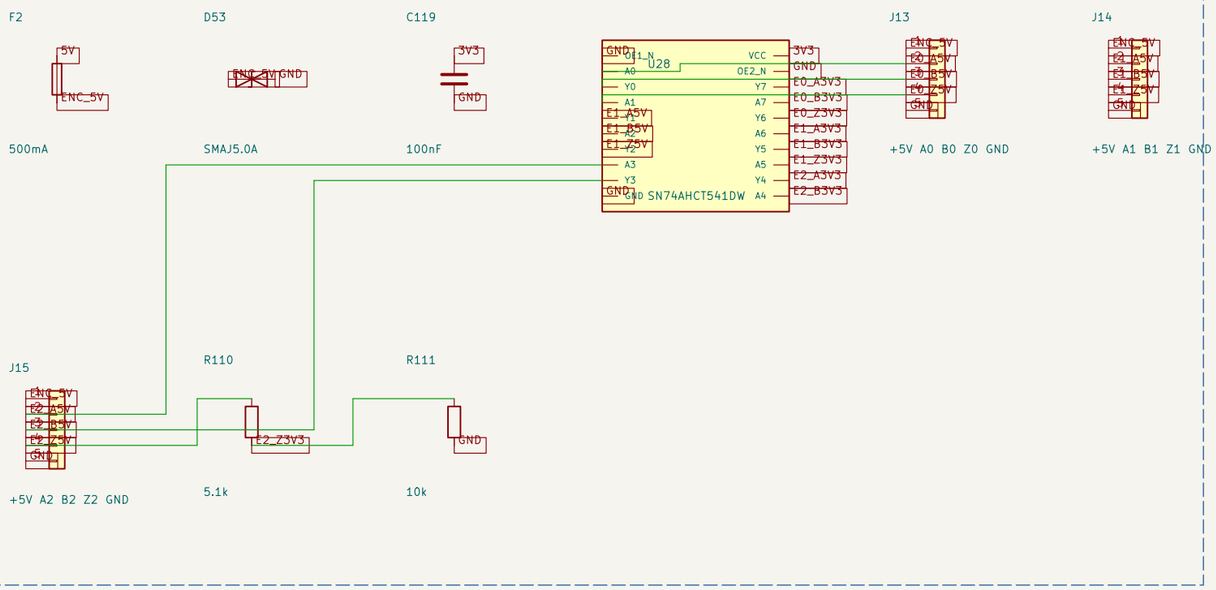
Sheet: /
File: display-python.kicad_sch

Title: LVDS transmitter, termination and connector

Size: A3	Date: 2026-11-03	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/1

Encoder inputs and 5 V interface | 9 components | 22 local nets

ENCODER POWER AND SIGNAL INTERFACE



Existing symbols grouped by function; local nets routed where practical

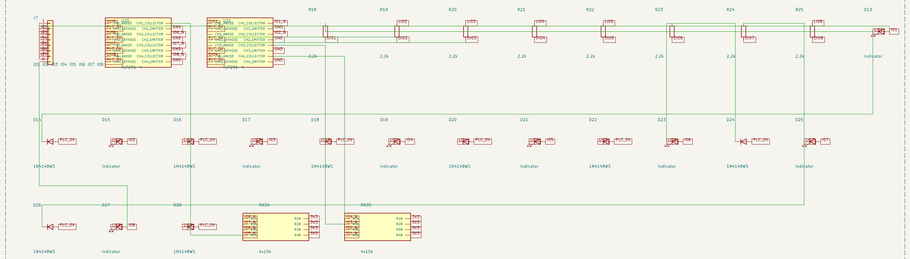
LeanCam / uCNC Python hierarchy experiment

Sheet: /
File: encoders-python.kicad_sch

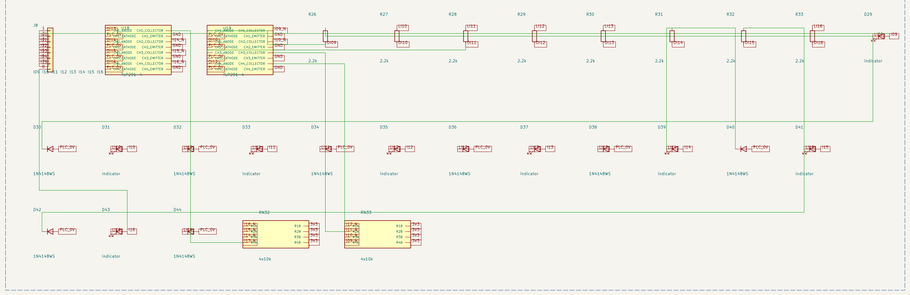
Title: Encoder inputs and 5 V interface

Size: A3	Date: 2026-11-03	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/1

PLC INPUTS I01–I08



PLC INPUTS I09–I16



E-STOP INPUT AND INDICATION

Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

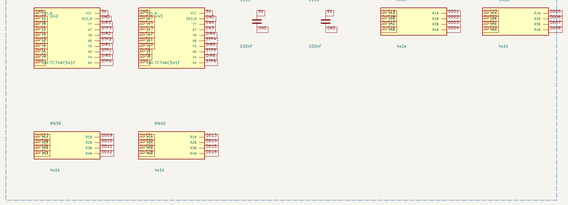
File: Field_Inputs--python.kicad_sch

Title: Isolated PLC inputs and E-stop

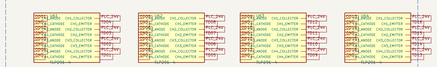
Size: A0	Date: 2026-11-03	Rev:
KiCad E.D.A. 10.0.1		id: 1/1

PLC outputs, motion interfaces and servo feedback | 72 components | 130 local nets

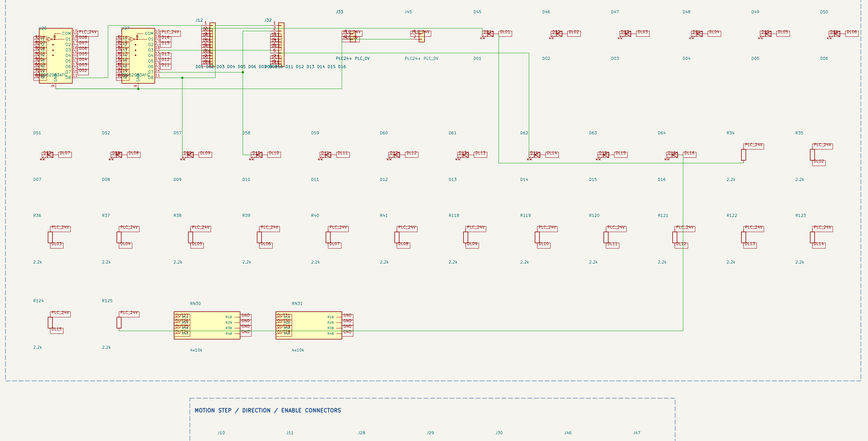
5 V OUTPUT BUFFERS AND SERIES RESISTORS



16-CHANNEL OUTPUT OPTOISOLATION



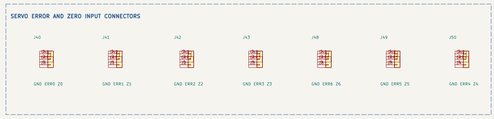
GENERAL OUTPUT DRIVERS AND CONNECTORS



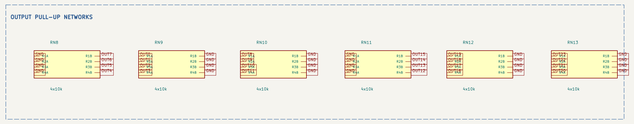
MOTION STEP / DIRECTION / ENABLE CONNECTORS



SERVO ERROR AND ZERO INPUT CONNECTORS



OUTPUT PULL-UP NETWORKS



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

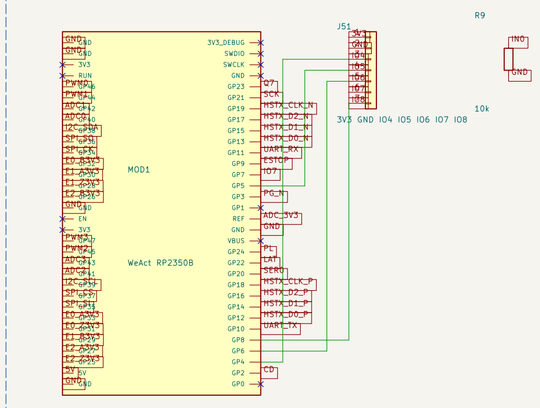
File: fpid_outputs-python.kicad_sch

Title: PLC outputs, motion interfaces and servo feedback

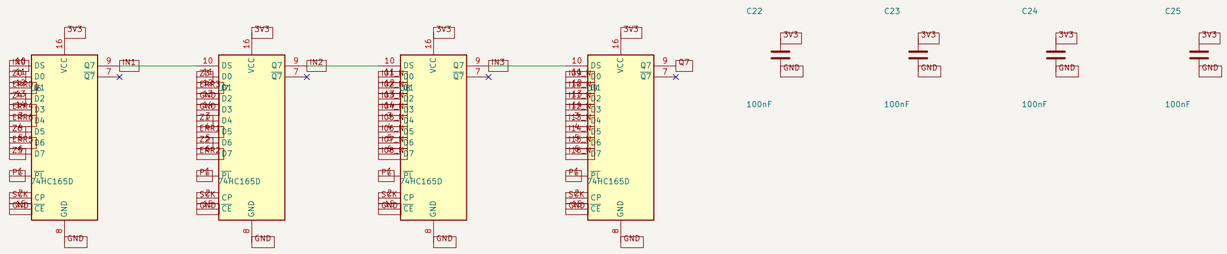
Size: A0	Date: 2026-11-03	Rev:
KiCad E.D.A. 10.0.1		Id: 1/1

RP2350B module, shift-register chains and safety I/O | 19 components | 119 local nets

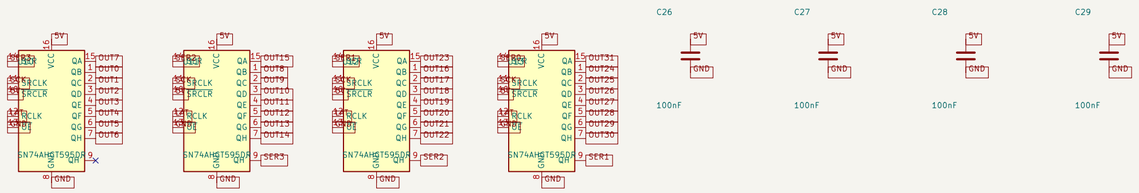
WeAct RP2350B AND EXPANSION HEADER



74HC165 INPUT SHIFT CHAIN



SN74AHCT595 OUTPUT SHIFT CHAIN



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

File: mcu_io-python.kicad_sch

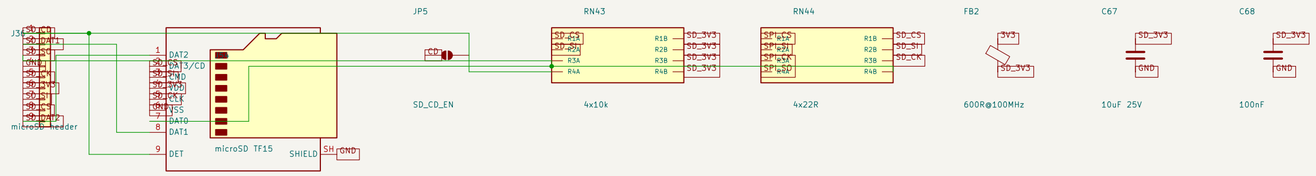
Title: RP2350B module, shift-register chains and safety I/O

Size: A2 | Date: 2026-11-03 | **Rev:**

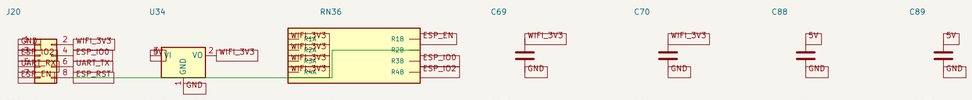
KiCad E.D.A. 10.0.1 | Id: 1/1

microSD, Wi-Fi and I2C interfaces | 16 components | 25 local nets

MICROSD SOCKETS AND SERIES TERMINATION



ESP-01 / ESP-01S SOCKET AND LOCAL REGULATOR



I2C SERVICE PORT



Existing symbols grouped by function; local nets routed where practical

LeanCam / uCNC Python hierarchy experiment

Sheet: /

File: storage_wifi_ui-python.kicad_sch

Title: microSD, Wi-Fi and I2C interfaces

Size: A2	Date: 2026-11-03	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/1